Practice workbook · Calculate BMI and place it in a weight category

Body mass index from mass and height and reading its category from a table.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-BMI-CALCULATE-CLASSIFY

1. Question 3 · part b.i · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part b.i only.

3 (a) In England, 3 319 266 people have diabetes.

(i) Give this value to 3 significant figures.

(1)

people with diabetes = ...

(ii) In the rest of the United Kingdom, 600 239 people have diabetes.

Give this value in standard form.

(2)

people with diabetes = ...

(iii) Diabetes can be caused by the inability of an organ to produce insulin.

Name the organ that produces insulin.

(1)

.....

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1. Question 3 · part b.i · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part b.i only.

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(b) The body mass index (BMI) is a measure that uses height and body mass to work out if the body mass is healthy.

The table shows the BMI values and the categories into which a person can be placed.

BMI	BMI category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25.0 to 29.9	overweight
30.0 to 39.9	obese
40.0 and over	morbidly obese

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(i) Calculate the BMI of a person with a body mass of 120 kg and a height of 1.95 m.

Use the equation

$$\text{BMI} = \frac{\text{body mass}}{\text{height}^2}$$

(2)

BMI = ...

(ii) Determine the BMI category that this person belongs to.

(1)

.....

.....

.....

.....

1. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

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2. Question 5 · part b.i · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.i only.

5 A teacher wants to calculate the body mass index (BMI) of a number of students.

(a) The teacher measures the height of students from four age groups.

There are two male and two female students in each age group.

The table shows the data collected.

Age in years	Height of student in cm				Mean height of students in cm
	Male		Female		
12	146.2	148.9	142.8	144.2	145.5
14	165.7	166.4	161.9	164.0	164.5
16	175.9	178.5	166.3	167.6	
18	180.9	181.3	171.2	174.3	176.9

(i) Calculate the mean height of the students aged 16.

Give your answer to one decimal place.

(2)

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mean height = ... cm

(ii) State two conclusions that the teacher could make from the data in the table.

(2)

1 .....

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2 .....

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(iii) Give two ways the teacher could improve his investigation.

(2)

1 .....

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2 .....

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2. Question 5 · part b.i · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.i only.

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(b) The table shows data for two different students.

Student	Height in m	Mass in kg	BMI
X	1.69	62	21.7
Y	1.46	71	

(i) BMI is calculated using the equation

$$\text{BMI} = \frac{\text{mass}}{\text{height}^2}$$

Calculate the BMI of student Y.
[mass measured in kg, height measured in m]

(2)

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BMI of student Y = ...

(ii) BMI can be used to assess whether a person has a healthy mass for their height.

BMI	Classification
below 18.5	underweight
18.5 – 24.9	normal weight
25 – 29.9	overweight
30 – 40	obese
above 40	morbidly obese

Using information from the table, explain how the BMI of student Y could affect her health.

(2)

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(Total for Question 5 = 10 marks)

2. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

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3. Question 3 · part b.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.ii only.

3 (a) In England, 3 319 266 people have diabetes.

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people with diabetes = ...

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(iii) Diabetes can be caused by the inability of an organ to produce insulin.

Name the organ that produces insulin.

(1)

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3. Question 3 · part b.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.ii only.

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Use the equation

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BMI = ...

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(1)

.....

.....

.....

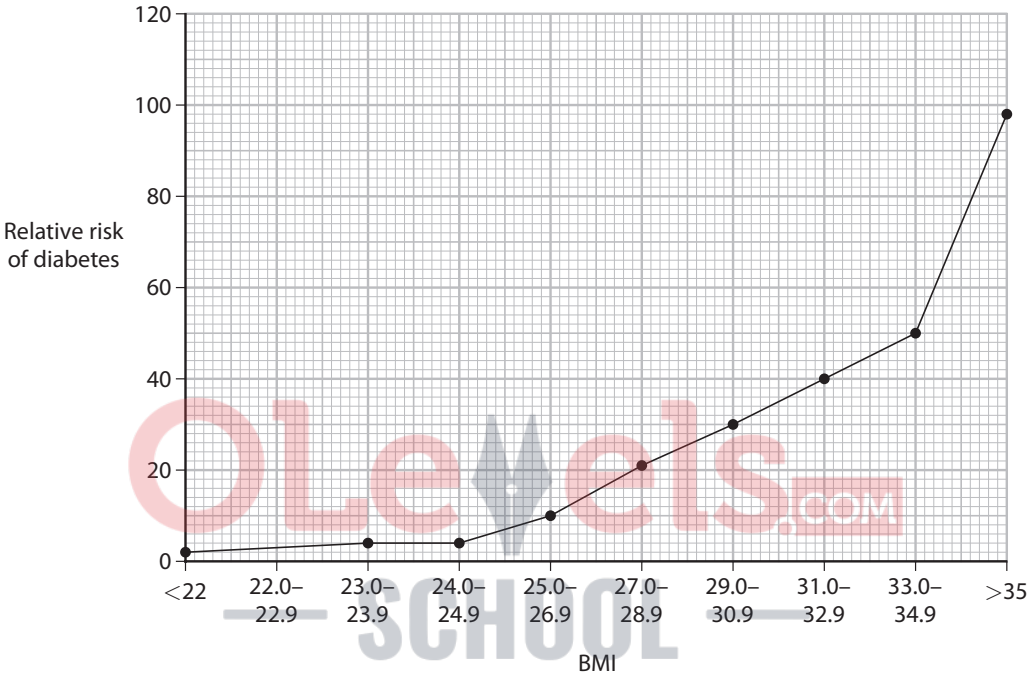
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3. Question 3 · part b.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.ii only.

(c) The graph shows the relative risk of developing a form of diabetes called type 2 for different BMI values.



(Source: adapted from Colditz GA, et al., *Ann Intern Med.*, 1995; 122:481–486)

(i) Describe the trend shown by the graph.

(2)

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3. Question 3 · part b.ii · 1 mark · Working and answer

Working and answer

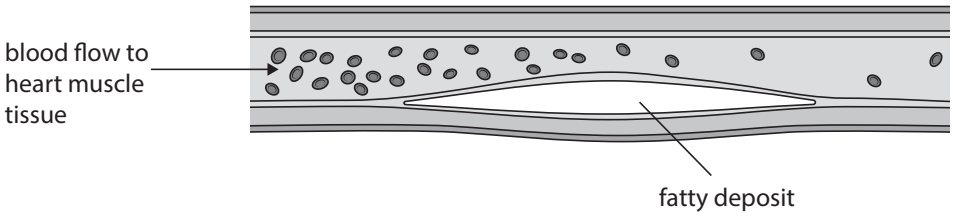
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4. Question 7 · part b.ii · 3 marks

June 2024 Paper 2

Further practice · Attempt part b.ii only.

7 (a) The diagram shows a fatty deposit in an artery carrying blood to the heart muscle.



(Source: adapted from https://www.heartandstroke.ca/heart-disease/conditions/atherosclerosis)

Explain how a build-up of a fatty deposit in an artery carrying blood to the heart muscle can cause heart disease.

(4)

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4. Question 7 · part b.ii · 3 marks

June 2024 Paper 2

Further practice · Attempt part b.ii only.

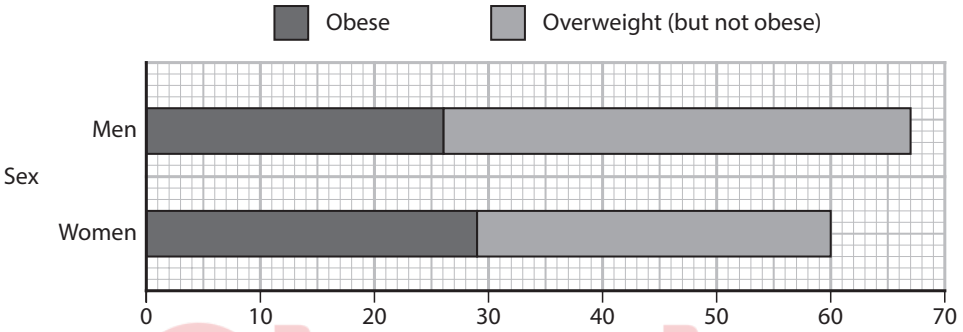
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(b) People with obesity are more at risk of heart disease than those who are overweight.

The chart shows the percentage of men and women in the United Kingdom who were either overweight or obese in 2018.



(Source: https://digital.nhs.uk/data-and-information/publications/statistical/health-survey-for-england/2018/summary)

(i) In 2018 there were 66.5 million people living in the UK. Of these people, 33 million were men.

Calculate how many more women than men were classified as obese.

(3)

extra number of women = ...

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P 7 4 6 3 0 A 0 1 9 2 4

Turn over ▶

4. Question 7 · part b.ii · 3 marks

June 2024 Paper 2

Further practice · Attempt part b.ii only.

(ii) Describe a calculation that could be used to determine if a person is obese.

(3)

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4. Question 7 · part b.ii · 3 marks

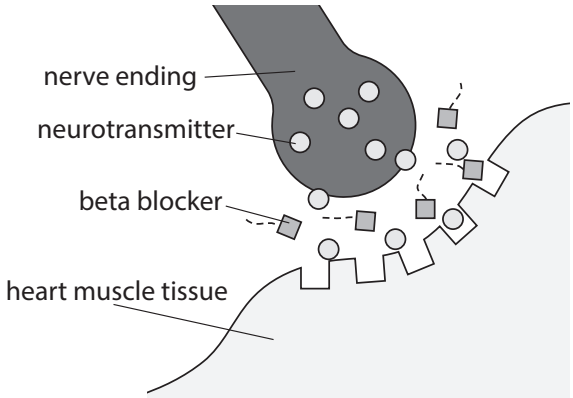
June 2024 Paper 2

Further practice · Attempt part b.ii only.

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(c) The diagram shows a junction between a neurone and heart muscle.



(Source: Adapted from: https://study.com/academy/lesson/what-are-beta-blockers-uses-types-examples-side-effects.html)

Use the diagram to explain how beta blockers can reduce the risk of heart disease.

(3)

(Total for Question 7 = 13 marks)

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4. Question 7 · part b.ii · 3 marks · Working and answer

Working and answer

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5. Question 4 · part b.i · 2 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

4 Carbohydrates are large molecules found in many body cells.

(a) Glycogen is a carbohydrate.

(i) Describe how the liver produces glycogen.

(2)

(ii) Name the hormone that stimulates the liver to produce glycogen.

(1)

(iii) Name a hormone that stimulates the liver to increase blood sugar levels.

(1)

(b) Consuming a diet high in carbohydrates can increase the BMI of a person.

The table gives information on the BMI categories of individuals.

BMI	Category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25 to 29.9	overweight
30 to 34.9	obese
35 to 39.9	severely obese
more than 39.9	morbidly obese

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5. Question 4 · part b.i · 2 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

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(i) A person has a height of 1.69 m and a mass of 67.0 kg.

Calculate the BMI of this person.

(2)

BMI .....

(ii) Give the BMI category of this person.

(1)

.....

(c) Obesity is a risk factor for heart disease.

Explain how obesity can cause heart disease.

(2)

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(Total for Question 4 = 9 marks)

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Working and answer

6. Question 4 · part b.ii · 1 mark

November 2024 Paper 1

Further practice · Attempt part b.ii only.

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November 2024 Paper 1

Further practice · Attempt part b.ii only.

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Working and answer